1 FIG-219 — A2 grouped retraining and overlap audit

Group: REV1-A2 **Evidence chain:** compact-input reconstruction **Status:** partial; provisional DFT anchor

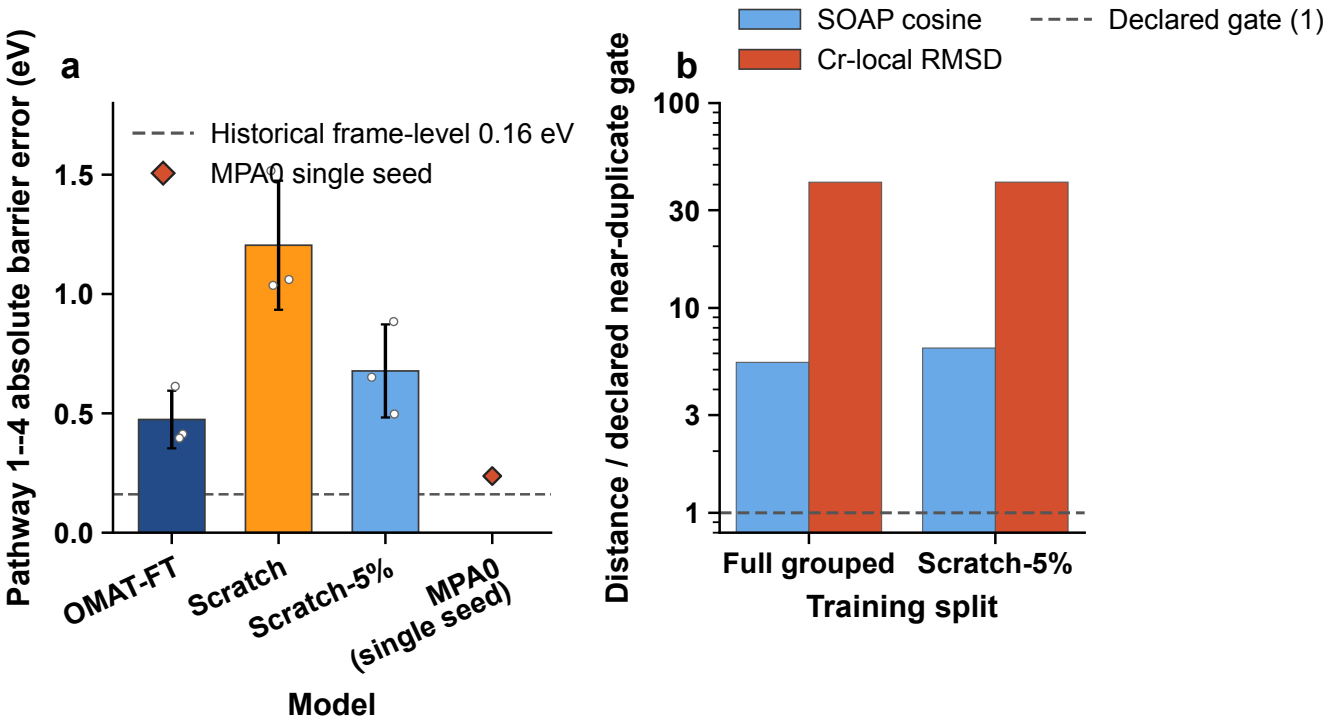


Figure 1: **Leakage-controlled grouped retraining and overlap audit for A2.** (a) Pathway 1–4 absolute fixed-path barrier errors for OMAT-FT, Scratch, and Scratch-5% under the trajectory-grouped protocol. Open points are seeds 123/234/345, bars/whiskers are mean ± sample standard deviation, and MPA0 is a single-seed comparator. The dashed line marks the historical frame-level 0.16 eV point estimate, which is not reproduced by the grouped FT-600K evaluation. (b) Minimum training-set SOAP and comparable Cr-local RMSD distances normalized by the declared near-duplicate gates. The dashed line at one is the declared near-duplicate gate; values above one lie outside that gate. RMSD is restricted to species-composition-compatible local environments, while SOAP covers all audited target images.

Results text draft. The grouped three-seed pathway 1–4 absolute barrier errors are 0.47 ± 0.12 eV for FT-600K (OMAT-FT start), 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for Scratch-5%. The single-seed MPA0 comparator is 0.24 eV. The full grouped and Scratch-5% splits show no detected exact or near-duplicate overlap under the declared metrics. **Discussion and boundary.** This reconstruction supports the leakage-controlled in-gap comparison and seed-level uncertainty. It does not establish that the provisional 0.34 eV DFT candidate is a final converged reference, does not make a claim about the unavailable foundation pretraining corpus, and does not rank the higher-distortion pathways. **Article placement.** Supporting Information, "MACE Model Performance Evaluation by RMSE" and "In-Gap Diffusion"; reviewer response A2. **Reviewer linkage:** overall status PARTIAL.

- **REV-A2 (PARTIAL)**: *Completed:* trajectory-grouped split, 25-image exact/SOAP/RMSD audit, and grouped FT-600K retraining show no detected overlap under the declared metrics and do not reproduce the historical 0.16 eV point estimate. *Still required:* close the independent QE convergence gate for the 0.34 eV reference candidate.
- **REV-A3 (PARTIAL)**: *Completed:* three-seed in-gap barrier uncertainty is shown. *Still required:* path-resolved deep-penetration references and classifications before a pooled OOD ranking.
- **REV-A5 (DONE for the plotted scope)**: the three-seed Scratch-5% control is included with its RMSE and pathway 1–4 barrier error.

Evidence pointers.

- Data/processed: `paper/response_to_reviewers/reviewer_a2_grouped_retraining/data/grouped_seed_comparison.csv`

- Data/processed: paper/response_to_reviewers/reviewer_a2_grouped_retraining/data/grouped_neb_barrier_summary.csv
- Data/processed: paper/response_to_reviewers/reviewer_a2_grouped_retraining/data/overlap_full_leakage_audit.json
- Data/processed: paper/response_to_reviewers/reviewer_a2_grouped_retraining/data/overlap_scratch5_leakage_audit.json
- Plot/processing script: paper/response_to_reviewers/reviewer_a2_grouped_retraining/plot_a2_grouped_retraining_audit.py
- Plot Atlas primitives: /Users/alina/Documents/Codex/plot_library/src/plot_atlas/charts.py

Machine-collected evidence candidates (generated from the A2 compact-input bundle; do not edit this file).

- **Caption/discussion source:** /Users/alina/Project/migrationbench-paper-repro/paper/response_to_reviewers/reviewer_a2_grouped_retraining/README.md
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/paper/response_to_reviewers/reviewer_a2_grouped_retraining/data/grouped_seed_comparison.csv
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/paper/response_to_reviewers/reviewer_a2_grouped_retraining/data/overlap_full_leakage_audit.json
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/paper/response_to_reviewers/reviewer_a2_grouped_retraining/data/overlap_scratch5_leakage_audit.json
- **Plot or processing script:** /Users/alina/Project/migrationbench-paper-repro/paper/response_to_reviewers/reviewer_a2_grouped_retraining/plot_a2_grouped_retraining_audit.py

Audit gap / next action: The plot is a reconstruction from compact derived evidence. The barrier comparison remains anchored to the provisional 0.34 eV DFT candidate until the independent QE convergence gate closes; higher-distortion pathways remain a validation gap and are not pooled.